

EEGNet Failure Analysis & Beyond

Defence presentation — PhysioNet EEGBCI motor-imagery subset

Slide 1 — The question and the setup

EEGNet works in the lab. **Where does it fail when you ask it to read a new person, and what does the failure tell us about how to fix it?**

Train	Test	Task	Channels	Sampling	Trials/subject
Subj 1–8	Subj 9–10	Imagined fists vs feet (PhysioNet Task 4)	64	160 Hz	~45

Three seeds (1337, 2024, 7). I always report **best AND worst** — no cherry-picking. Source of truth: `results/metrics.json`.

Slide 2 — Three experiments to localise the failure

Setting	Train	Test
Cross-subject, full pool	1–8 (8 subjects)	9–10
Cross-subject, reduced pool	1, 2, 3 (3 subjects)	9–10
UMAP of the model's internal features	diagnostic	diagnostic

No within-subject baseline. The thing under test is **transfer to a new person**, not raw capacity. Within-subject EEGNet is already known to work (Lawhern 2018).

Slide 3 — Baseline results — the real clue isn't in accuracy

Setting	Best	Worst	Sil(class)	Sil(subject)
Full 8 subj	0.644	0.511	0.010	0.084
Reduced 3 subj	0.656	0.533	-0.002	0.062

Two things to notice:

1. **More training subjects did NOT help.** 8 vs 3 gives the same accuracy on unseen people ($0.644 \approx 0.656$). So one more subject is worth roughly zero — this isn't a "need more data" story.
2. **Subject clustering is $\sim 8\times$ class clustering** (0.084 vs 0.010).
The model's primary internal feature is "which person is this?", not "fists or feet?".

Sil(subject) = how strongly the model's internal features group

Slide 4 — Why this happens — the mechanism

EEGNet's spatial filter is a weight matrix of shape `(filters, channels)` — one weight per (filter, electrode-index) pair. It addresses electrodes by **index** (slot number), not by **physical position** on the head.

Two structural facts:

1. The depthwise weight has **no notion of physical electrode position**. Move the cap 1 cm and the same weight is now reading a different brain region — and the network has no way to know.
2. Each person's electrical fingerprint (skull thickness, electrode contact, head shape) shows up in the resting covariance with very high signal-to-noise. The classifier latches onto it because it's *easier to learn than the actual class signal*.

Slide 5 — Part 2 — TopoNet — two changes

5a. Per-trial Riemannian whitening

$$\tilde{X} = (XX^\top / T + \varepsilon I)^{-1/2} X$$

In plain terms: rescale each trial so its covariance matrix becomes the identity. This mathematically subtracts out the person-specific covariance fingerprint. No calibration data needed from the test subject — every trial gets whitened on its own.

5b. Coordinate-Conditioned Spatial Filters (CCSF)

$$W[c, k] = \text{MLP}([\text{pos}_c, e_k])$$

The spatial filter is no longer a lookup table. It's a small network that takes each electrode's actual 3D position on the head as input and outputs the filter weight for it. Same physical

Slide 6 — TopoNet results — an honest split

Setting	Best	Worst	Sil(class)	Sil(subject)
Baseline, 8 subj	0.644	0.511	0.010	0.084
TopoNet, 8 subj	0.489	0.456	0.005	0.014
Baseline, 3 subj	0.656	0.533	-0.002	0.062
TopoNet, 3 subj	0.622	0.600	0.006	0.028

What's working and what isn't:

- **Mechanism works.** Subject clustering drops 6× on the full pool and 2.3× on the reduced pool — the Part 1 diagnostic moves a lot.
- **Full-pool accuracy drops.** TopoNet sits below the baseline here

Slide 7 — Why TopoNet wins on 3 subjects and loses on 8

The motor-imagery class signal for runs 6/10/14 lives in the **lateral-vs-medial covariance contrast** — bilateral hand area (C3/C4) for imagined fists, midline foot area (Cz) for imagined feet.

Per-trial whitening flattens the covariance across the board, so it strips the class signal along with the person-specific component — they live in the same algebraic object.

- **Full pool:** EEGNet already has enough data to find robust filters on its own. Whitening becomes *cost without benefit*.
- **Reduced pool:** EEGNet was overfitting. Whitening's prior gives it a cleaner objective and stabilises every seed.

I threw the baby out with the bathwater on the full pool. The

Slide 7b — Ablation breaks TopoNet into its two halves

`scripts/ablation_part2.py` :

Config	8-subj best/worst	3-subj best/worst	sil(subj) 8 / 3
Baseline	0.644 / 0.511	0.656 / 0.533	0.084 / 0.062
Whitening only	0.600 / 0.489	0.600 / 0.489	0.022 / 0.017
CCSF only	0.522 / 0.422	0.611 / 0.567	0.022 / 0.020
TopoNet (both)	0.489 / 0.456	0.622 / 0.600	0.014 / 0.028

Three findings:

1. **Each half alone cuts subject clustering $\sim 4\times$.** Both halves independently hit the mechanism from Part 1.
2. **They stack badly on accuracy** — both remove some of the same

Slide 8 — Part 3 — Hemispheric contrast channels

The novel idea. Half-page rationale in `report/part3_idea.md` .

Claim: the useful signal in EEG often sits in the *relationships* between electrodes, not in any one channel's amplitude. Volume conduction (the skull and scalp spread each brain signal across many electrodes) means 64 channels are really 64 overlapping views of a much smaller set of underlying sources. The relationships are already in the input, just buried.

Idea: add 10 extra channels next to the raw 64. Nine are left-right symmetric pairs near motor cortex (C3 – C4, FC3 – FC4, CP3 – CP4 and neighbours). The tenth is the hand-vs-foot contrast: $(C3 + C4)/2 - Cz$. Input becomes 74 channels.

Add, not replace. If the contrasts are useless the model can

Slide 9 — Why this is grounded, specific, and (I think) not just bipolar

Grounded. EEG has high time resolution but coarse spatial resolution. Volume conduction means neighbouring channels mostly carry the same thing. Useful information lives in the differences between electrodes.

Specific. Ten exact contrasts, named electrodes, fixed subtraction. `scripts/plot_contrasts.py` lets a reviewer verify the C3 – C4 class-conditional sign flip themselves.

Not just a bipolar montage. Bipolar, surface Laplacian, common-average, CSP, ICA — they all *replace* the raw signal with a fixed transform before the network sees the data. Once that's done, the raw channels are gone.

Slide 10 — Sanity check on the engineered feature

`scripts/plot_contrasts.py` → `results/figures/part3_contrasts.png` .

Trial-averaged contrast traces, split by class:

- **C3 – C4 shows a sign flip between classes** (T1 mean negative, T2 mean positive) — the lateralization pattern the contrast was built to pick up.
- Effect size is modest at the trial level (Cohen's $d \approx 0.14$ on $C3 - C4$, ≈ 0.05 on $(C3 + C4)/2 - Cz$). The contrasts are class-informative but **not classifiers on their own** — they're a useful representation for the network to lean on, not a model.

Slide 11 — Part 3 results — the prediction held on the bigger pool

Setting	Best	Worst	Sil(class)	Sil(subject)
Baseline, 8 subj	0.644	0.511	0.010	0.084
EEGNet + contrasts, 8 subj	0.667	0.556	-0.001	0.037
Baseline, 3 subj	0.656	0.533	-0.002	0.062
EEGNet + contrasts, 3 subj	0.533	0.511	0.005	0.037

Full 8-subject pool — hypothesis held. Subject clustering drops $2.3\times$ ($0.084 \rightarrow 0.037$). Best accuracy up ($0.644 \rightarrow 0.667$). Worst case up ($0.511 \rightarrow 0.556$). Mechanism and outcome both lined up with what I'd predicted.

Reduced 3-subject pool — mechanism still works, but accuracy collapses. Subject clustering still drops ($0.062 \rightarrow 0.037$), so

Slide 11b — Did the model actually use the contrasts? — I checked

`scripts/part3_filter_analysis.py` → `results/figures/part3_filter_weights.png`.

I ran the test. The numbers came in mixed.

- Random baseline (every channel weighted equally): $10/74 \approx 0.135$.
- Trained mean per-filter share on contrasts: **0.150** — only $1.11\times$ the baseline. Small shift in aggregate.
- Per-filter spread: **0.031 to 0.263**. Five filters at 0.20–0.26 (so $1.5\text{--}2\times$ baseline). Five at 0.03–0.10 (basically ignoring the contrasts). The other six near the baseline.

So I'm walking the strong reading back. The aggregate number doesn't support "the model heavily leans on the prior" — I'd be overclaiming if I said that

Slide 12 — Defence Q&A I'm expecting

"Isn't this just a bipolar montage / surface Laplacian? Those are decades old."

Bipolar and Laplacian replace the raw signal with a fixed transform before the network sees anything — once it's done, the raw channels are gone. What I do is different: I leave the raw 64 in place and add 10 contrast channels next to them, and the depthwise filter decides per filter how much to weight each side. I ran the test on myself

(`scripts/part3_filter_analysis.py`) — trained mean share on contrasts came back at 0.150 against a 0.135 random baseline, only ~11% above chance. I'm not going to claim the model heavily leans on the prior — the aggregate number doesn't back that. But the per-filter spread is wide (0.031 to 0.263): some filters

Slide 13 — What I would build next

1. **Subject-conditional whitening** — strip only the subject-specific covariance, keep the class-discriminative asymmetry that simple whitening also wipes out.
2. **Frozen contrast weights on small pools** — freeze the depthwise weights on the 10 contrast channels in the reduced-data regime, so the prior is preserved without paying the parameter cost.
3. **Pre-train CCSF MLP across multiple EEG datasets** — it's channel-count agnostic, so cross-corpus pre-training is free.

Slide 13b — Visual evidence — topomaps tell the same story

`results/figures/topomap_eegnet.png` — EEGNet's 16 depthwise filters.

Fragmented, noisy spatial profiles — no clean motor-cortex localisation. These aren't motor-area detectors; they're person-specific covariance patterns.

`results/figures/topomap_ccsf.png` — CCSF's 16 filters.

Smooth gradients, all nearly identical. The MLP-on-position prior collapses to a small set of low-frequency spherical patterns. This explains CCSF's full-pool underperformance: not too little capacity, but too little *diversity*.

`results/figures/per_subject_accuracy.png` — EEGNet does much better on S10 (0.71) than S9 (0.58). The 0.644 headline number averages over a real asymmetry — subjects are not equally

Slide 13c — Classical reference — CSP + LDA

Setting	best / worst
CSP+LDA, 8 subj	0.489 / 0.489
CSP+LDA, 3 subj	0.367 / 0.367

CSP sits **at chance** on the full pool and **below chance** on the reduced pool. CSP filters are explicitly person-specific covariance maximisers — under reduced-data cross-subject training they actively anti-align to the test subjects' covariance.

It's the same failure mechanism Part 1 spelled out for EEGNet, showing up much more starkly in a model that's covariance-based by construction.

EEGNet (0.570 mean) sits above CSP because it has

Slide 14 — Take-aways

- The cross-subject failure is **structural, not statistical**.
Adding subjects doesn't fix it ($0.644 \approx 0.656$). CSP+LDA sitting at chance independently confirms the mechanism.
- The diagnostic that reveals the mechanism is **subject clustering**, not accuracy. Accuracy on a 90-trial test set has a $\pm 5\%$ noise floor.
- **Part 2 ablation** isolated CCSF as the cleaner half — best worst-case test on the reduced pool (0.567). Topomaps show CCSF's next-step problem (filter collapse).
- **Part 3 hemispheric contrasts** validated the inductive-bias hypothesis on the full pool (subject clustering $\downarrow 2.3\times$, best test $\uparrow 0.644 \rightarrow 0.667$, worst $\uparrow 0.511 \rightarrow 0.556$). On the small pool the mechanism still works but the parameter cost wins